Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738157048252-Chapter 8 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

INTRODUCTION

In this unit, you'll take everything you learned in Unit 1 and start creating your very own snoring detector! This is where you'll plan, design, and begin building your device. You'll choose the right tools and parts (like MODI modules), figure out how your device will work, and start bringing your ideas to life.

BRAINSTORMING IDEAS

The Smart Posture Detector we are going to build will use sensors to monitor your distance from the monitor and give you feedback when you are too close. Here's how it works:

1. THINK ABOUT WHAT YOUR DEVICE NEEDS TO DO:

Your snoring detector needs to be able to "hear" when someone is snoring. How might you use technology to listen for sounds during sleep?

2. WHAT SENSORS COULD YOU USE?

What kinds of sensors would help your device detect snoring? Think about how sound sensors work and how they might be used in your design.

3. HOW WILL YOUR DEVICE RESPOND?

Once your device detects snoring, what should it do? Should it alert the person to wake up? How can it do that without disturbing others?

TIPS

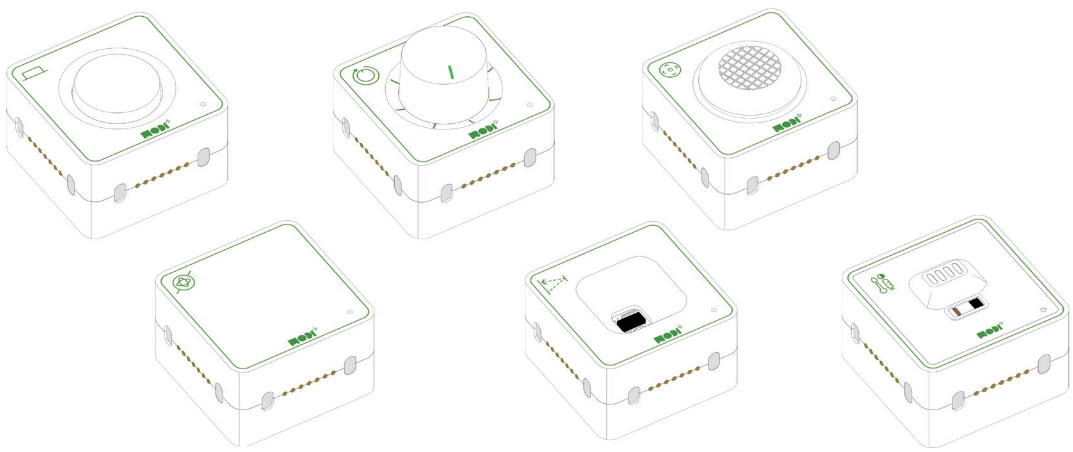
Think about the main parts of your Snoring Detector: a sensor to detect snoring, a way to process this, and an alert system. What MODI modules could you use for these parts? Write down your ideas.

Required Modules & Brainstorming

To build your snoring detector, we need to decide which MODI modules to use. Here are some ideas to get started:

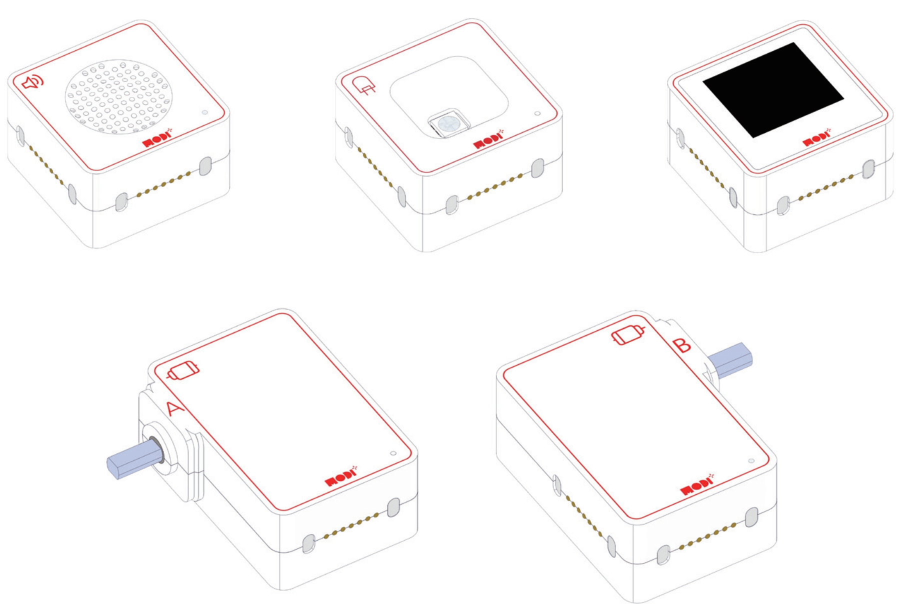
SENSOR MODULES

We need a sensor that can detect sound. Think about how different sensors could 'hear' snoring. Which sensor would work best for this purpose?



OUTPUT MODULES

Once the snoring is detected, how will the person be alerted? Consider using lights or sounds. Which output module will best grab their attention without waking others?



Required Modules & Brainstorming

Which MODI modules do you need for the creation?

DIAL

BUTTON

ENVIRONMENT

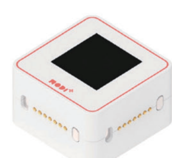
JOYSTICK

TOF

SPEAKER

IMU

LED

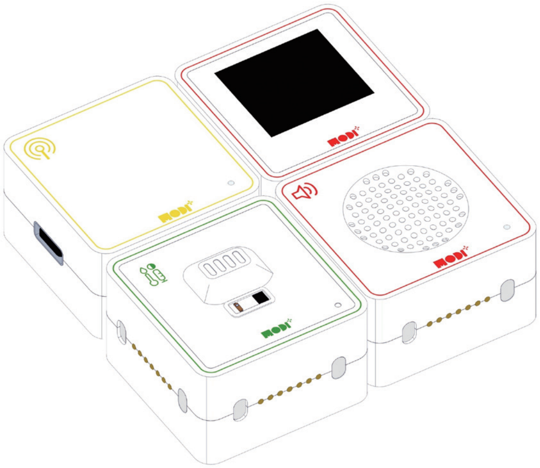


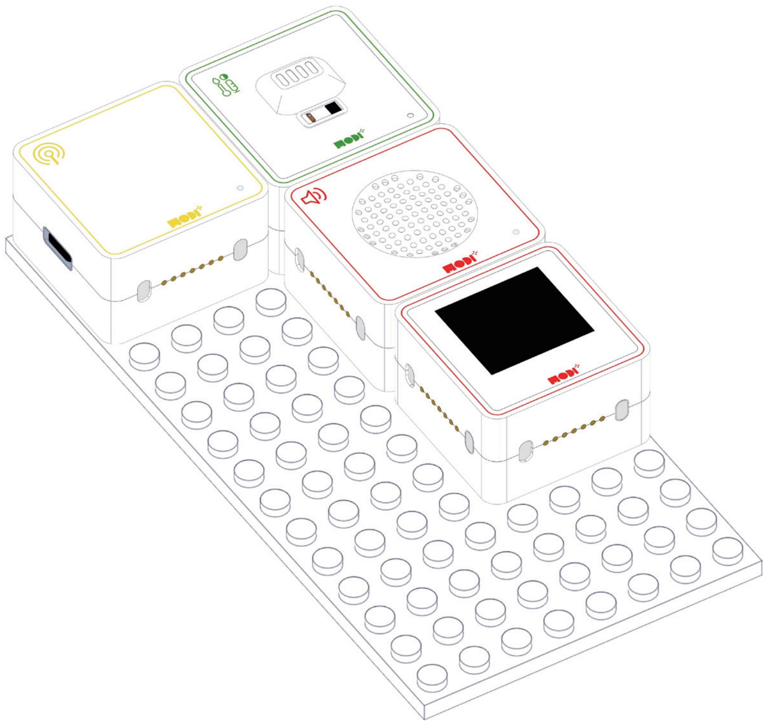
DISPLAY

MOTOR A

MOTOR B

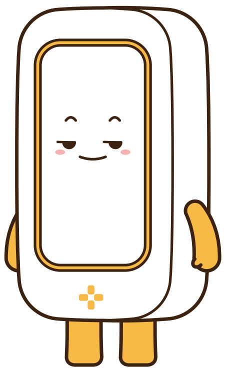
WHEN WE COMBINE ALL THE MODULES, THE CREATION HAS TO BE LOOK LIKE THIS.





TIPS

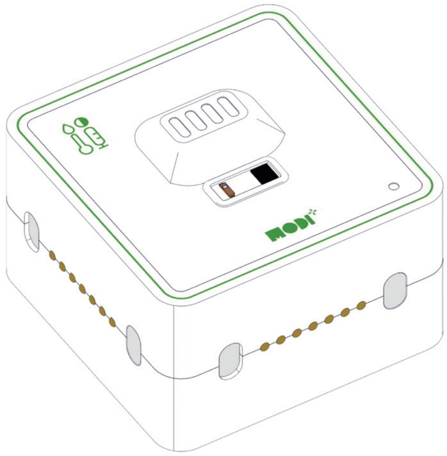
MODI modules can be connected in any direction! This makes building your projects easier and more flexible. No need to worry about which way they face!



Module Functionality

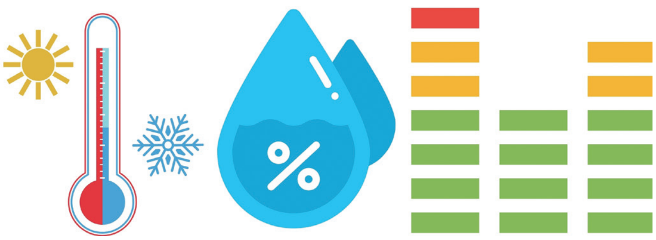
ENVIRONMENT SENSOR

The Environment Sensor is like the nose and ears of your snoring detector. It can "sense" different things like temperature, humidity, and even the air quality around you. Here's how it works:



1. SENSING THE ENVIRONMENT:

The sensor collects information about the air around it, like how warm or cool it is, how much moisture is in the air, and even how loud it is.



2. DETECTING CHANGES:

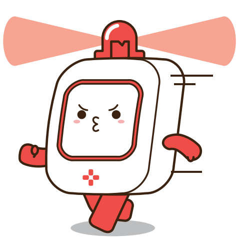
If the sensor detects something unusual, like a big change in temperature or humidity, it sends this information to your snoring detector.

3. SENDING INFORMATION:

The sensor sends the information it has collected to the other modules in your device. This helps your snoring detector decide what to do next.

4. HELPING YOUR DEVICE RESPOND:

Based on the information from the Environment Sensor, your snoring detector can decide if it needs to alert you with a light, sound, or vibration.

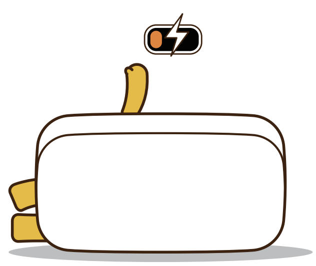


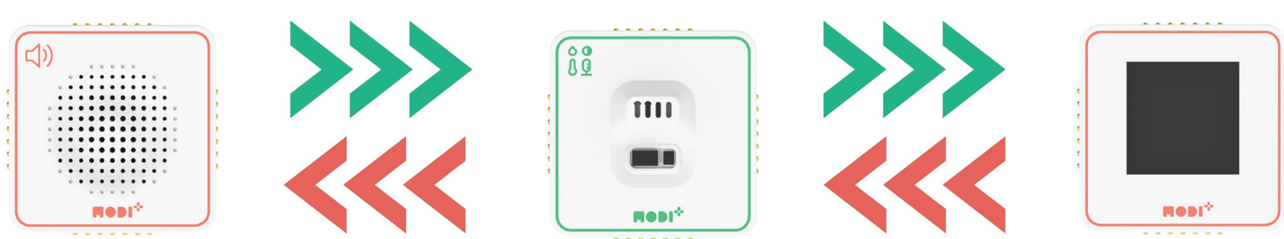
WHY THE ENVIRONMENT SENSOR IT USEFUL IN A SNORING DETECTOR?

The ToF sensor is used in many cool ways because it can measure distance very well. Here are some places where you might find it:

DETECTS SNORING SOUNDS:

The Environment Sensor can "hear" the sound of snoring by picking up noise levels in the room. When the sensor detects snoring, it sends this information to your snoring detector.





PROVIDES ACCURATE FEEDBACK:

By accurately detecting the sound of snoring, the sensor helps your snoring detector respond quickly with an alert, like a light or sound, to help improve sleep quality.